Indian Institute of Technology, Madras - BS in Data Science and Applications

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name :	IIT M DEGREE AN EXAM QDB2 15 Mar 2026
Subject Name :	2026 Mar15: IIT M AN EXAM QDB2
Creation Date :	2026-03-05 10:37:28
Duration :	120
Total Marks :	637
Display Marks:	Yes
Share Answer Key With Delivery Engine :	Yes
Actual Answer Key :	Yes
Calculator :	Scientific
Magnifying Glass Required? :	No
Ruler Required? :	No
Eraser Required? :	No
Scratch Pad Required? :	No
Rough Sketch/Notepad Required? :	No
Protractor Required? :	No
Show Watermark on Console? :	Yes
Highlighter :	No
Auto Save on Console?	Yes
Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Group I

Group Number :	1
Group Id :	64065332737
Group Maximum Duration :	0
Group Minimum Duration :	90

Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	637
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No
Revisit allowed for group Instructions? :	Yes
Maximum Instruction Time :	0
Minimum Instruction Time :	0
Group Time In :	Minutes
Revisit Section :	Yes
Action on Revisit Section :	View and Edit
Navigate To Group Summary From Last Question? :	No
Disable Submit Button During Assessment? :	No
Section Selection Time? :	0
No of Optional sections to be attempted :	0

Speech technology

Section Id :	640653129732
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	3
Number of Questions to be attempted :	3
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653362928
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406531961550

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536373785. ✔ Yes

6406536373786. ✖ No

Sub-Section Number :	2
Sub-Section Id :	640653362929
Question Shuffling Allowed :	No

Question Number : 2 Question Id : 6406531961551

Correct Marks : 12

Question Label : Multiple Choice Question

1. (1 point) Which of the following best describes the role of the Mel-Frequency Cepstral Coefficients (MFCC) in speech processing?

   (a) They capture pitch information directly.
   (b) They represent the short-term power spectrum on a perceptual scale.
   (c) They measure phase variations.
   (d) They enhance the high-frequency components.

2. (1 point) The phoneme `/f/` is a _____.

   (a) glide
   (b) nasal
   (c) fricative
   (d) stop

3. (1 point) Short Time Fourier Transform (STFT) is required to analyze speech because speech is

   (a) quasi-stationary
   (b) stationary
   (c) non-stationary

4. (1 point) _____ membrane in the ear acts as a frequency analyzer.

5. (1 point) For perfect reconstruction of a bandlimited signal, the sampling frequency must be

   (a) equal to twice the maximum frequency
   (b) greater than or equal to twice the maximum frequency
   (c) lesser than or equal to twice the maximum frequency
   (d) none of the above

6. (2 point) What is the minimum sampling frequency of a signal that was low pass filtered with a cut-off frequency of 10kHz to avoid aliasing?

   Note: 1 mark for the right answer and 1 mark for explaining the formula

7. (2 point) When spectrograms are obtained using longer time-windows, we get _____ frequency-resolution.

   (a) good
   (b) poor

   Note: 1 mark for the right option and 1 mark for explanation

8. (3 points) Comment on the linearity and time-invariance of the following systems?

   (a) $y[n] = 2x[n] + 3$
   (b) $y[n] = x[n]^2$
   (c) $y[n] = x[2n]$

Options :

6406536373787. ✔ I have answered this question in the offline paper

6406536373788. ✖ I have skipped this question

6406536373789. ✖ This question does not apply to me

Sub-Section Number :	3
Sub-Section Id :	640653362930
Question Shuffling Allowed :	No

Question Number : 3 Question Id : 6406531961552

Correct Marks : 13

Question Label : Multiple Choice Question

9. (3 points) Consider a weather HMM with 2 hidden states: *Sunny (S)* and *Rainy (R)*. The transition probabilities are:

$$A = \begin{bmatrix} P(S \to S) & P(S \to R) \\ P(R \to S) & P(R \to R) \end{bmatrix} = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

The emission probabilities (observations are *Walk*, *Shop*, *Clean*):

$$B = \begin{bmatrix} P(\text{Walk}|S) & P(\text{Shop}|S) & P(\text{Clean}|S) \\ P(\text{Walk}|R) & P(\text{Shop}|R) & P(\text{Clean}|R) \end{bmatrix} = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

Initial probabilities:

$$\pi = [P(S) = 0.6,\ P(R) = 0.4]$$

Given the observation sequence $O = (\text{Walk}, \text{Shop}, \text{Clean})$, use the **Viterbi algorithm** to compute the most likely sequence of weather states.

10. (5 points) In speech recognition, Gaussian Mixture Models (GMMs) are used to model the probability distribution of feature vectors such as MFCCs.

Suppose you have a 1-dimensional MFCC feature $x = 2.0$. The feature is modeled using a **2-component GMM** with the following parameters:

- Component 1: $\mu_1 = 1.0,\ \sigma_1^2 = 1.0,\ w_1 = 0.6$
- Component 2: $\mu_2 = 3.0,\ \sigma_2^2 = 1.0,\ w_2 = 0.4$

(a) Compute the probability density $p(x)$ of x under this GMM.

(b) Compute the posterior probability that the feature x was generated by **Component 1**.

Hint: The PDF of a 1D Gaussian is

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$$

11. (5 points) We discussed two approaches for decoding in Hidden Markov Models (HMMs): the **greedy** approach and the **Viterbi** algorithm.

Consider a part-of-speech (POS) tagging problem modeled as a first-order HMM. Given the observed word sequence: *"Time flies fast"* and that the first word "Time" is tagged as a **Noun**, determine the most probable tag sequence for the next two words using:

(a) the greedy approach

(b) the Viterbi algorithm

Hint: In the Viterbi approach, keep track of the best path probabilities for each possible tag at each position.

Assume the following transition and emission probabilities:

Transition	To Noun	To Verb	To Adjective
From Noun	0.3	0.5	0.2
From Verb	0.4	0.4	0.2
From Adjective	0.2	0.3	0.5

Word	P(Word—Noun)	P(Word—Verb)	P(Word—Adjective)
flies	0.1	0.8	0.1
fast	0.2	0.3	0.5

Options :

6406536373790. ✔ I have answered this question in the offline paper

6406536373791. ✖ I have skipped this question

6406536373792. ✖ This question does not apply to me